

N KEERTHI VASAN

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Portfolio: <https://nvasank.github.io/>

PROFESSIONAL SUMMARY

Cybersecurity Engineer (In Training) | Cybersecurity · Post-Quantum Security · Quantum Computing (Interest) · Ethical Hacking · VAPT · IoT Security

M.Tech Cybersecurity (DIAT Pune, MoD) · B. Tech CSE with Specialisation in Cybersecurity (VIT 2025) · DRDO Intern

Cybersecurity engineer with a focused five-year academic background in the discipline — starting with a B. Tech specialization in Cybersecurity at VIT and continuing through M. Tech at the Defence Institute of Advanced Technology (DIAT), Pune, under the Ministry of Defence. Currently interning at DRDO Bhawan, New Delhi. Trained in offensive and defensive security operations, cryptography, digital forensics, network security, and cyber-warfare fundamentals. Brings hands-on experience in vulnerability assessment and penetration testing from a cybersecurity internship, practical IoT and embedded systems knowledge from three hardware-software projects, and a growing interest in quantum computing and its implications for national security and cryptographic systems. Selected to DIAT after clearing the institute's competitive entrance examination and ranking within the top 100 candidates nationally.

CORE COMPETENCIES

Ethical Hacking & VAPT • Vulnerability Assessment • Penetration Testing • Red/Blue/Purple Team Operations • Kali Linux • OWASP Top 10 • MITRE ATT&CK • Incident Response • Digital Forensics • SIEM Concepts • Threat Modelling • Network Security • Cryptography & Post-Quantum Security (NIST FIPS 203/204/205) • Quantum Computing (Active Interest) • IoT & Embedded Systems Security • RFID Access Control • Two-Factor Authentication • Python • TryHackMe

TECHNICAL SKILLS

Languages: Python, Java, C (Embedded), JavaScript, SQL, R, HTML/CSS, Bash

Security Tools: Kali Linux, Wireshark, Nmap, Burp Suite, Metasploit (Basics), John the Ripper (Basics), Aircrack-ng (Concepts), TryHackMe

Cryptography: NIST PQC Standards (ML-KEM/Kyber, ML-DSA/Dilithium, SLH-DSA/SPHINCS+), Lattice-Based & Hash-Based Cryptography, OpenSSL, liboqs (Open Quantum Safe)

Quantum Computing: Foundational concepts — qubits, superposition, entanglement, quantum gates, quantum key distribution (QKD); actively learning Qiskit and PennyLane frameworks

AI/ML (Developing): Scikit-learn, NumPy, Pandas, MATLAB, NLP Fundamentals

IoT & Embedded: Arduino, Embedded C, I2C Communication, Sensor Integration (Heartbeat, Thermal, Force/Impact), GPS Modules

Tools & Platforms: Git/GitHub, Linux, Docker (Basics), MySQL, Power BI, MS Excel, Arduino IDE

Frameworks & Standards: OWASP Top 10, NIST Cybersecurity Framework, ISO 27001 Awareness, MITRE ATT&CK, FIPS 203/204/205, Agile/Scrum

PROFESSIONAL EXPERIENCE

Cybersecurity Research Intern | DRDO Bhawan, New Delhi — Ministry of Defence, Govt. of India July 2026 – July 2027

Interning at the headquarters of India's premier defence research organisation under the Ministry of Defence, contributing to defence-grade cybersecurity and research work.

Cybersecurity Intern | TechnoHacks EduTech

Jan 2023 – May 2023

Worked within the organisation's security team conducting hands-on vulnerability assessment, simulated penetration testing, and incident-response documentation across network and web application infrastructure.

- **Vulnerability Assessment (Kali Linux / Nmap / Burp Suite):** Situation: Organisation lacked a documented, OWASP-aligned view of its security gaps. Task: Conduct a structured vulnerability assessment across network configurations, web applications, and access-control layers. Action: Used Kali Linux toolset including Nmap, Wireshark, and Burp Suite to identify and document security gaps mapped against OWASP Top 10. Result: Delivered a security gap report that reduced exposure across the organisation's top attack-vector categories.
- **Simulated Penetration Testing (Red Team Simulation):** Situation: Authentication flows had not been adversarially tested. Task: Execute simulated penetration tests on internal assets to surface exploitable misconfigurations. Action: Identified

authentication flow weaknesses through manual testing and contributed structured remediation recommendations. Result: Findings were adopted into the team's hardened security baseline.

- **Incident Response Documentation (Blue Team Collaboration):** Situation: The team lacked standardised incident-response procedures. Task: Design incident-response checklists and authentication protocols with a 5-member team. Action: Collaboratively documented response workflows and hardened authentication protocols. Result: Documentation was adopted into the internal security playbook.
- **Security Tool Benchmarking:** Situation: The team needed to select the right tools for ongoing monitoring. Task: Evaluate Wireshark, Nmap, and Burp Suite against operational requirements. Action: Conducted a structured comparative analysis. Result: Analysis directly informed the team's monitoring-tool selection.

KEY PROJECTS

Safe-X — IoT Smart Safety Helmet (Embedded Systems / IoT Security) | VIT *Aug 2023 – Oct 2023*

Situation: Road accident fatalities in India are significantly worsened by delayed emergency response. Task: Build a wearable safety system that detects accidents and alerts emergency contacts automatically. Action: Engineered an Arduino-based helmet integrating heartbeat (MAX30102), fall/impact, and thermal (DHT11) sensors, with I2C sensor communication, a tamper-resistant sensor data pipeline, and automated GPS-based SOS alerts. Result: Cut simulated emergency-response time by 75% in controlled testing; won VIT's university innovation competition and was selected as a department reference project for IoT security design.

MINDASSIST — AI-Assisted Mental Health Platform (NLP / Security Architecture) | VIT Capstone *Jun 2024 – Dec 2024*

Situation: Campus mental health support was understaffed and inaccessible outside narrow hours. Task: Design an AI-assisted platform offering early-stage support while protecting highly sensitive behavioural data. Action: Built an end-to-end platform applying NLP-based sentiment analysis for mood tracking, with a full security layer including encryption, role-based access control, and privacy-by-design data handling. Result: Demonstrated early-stage distress-detection logic on a sample group; responsible for the complete security architecture of the system.

DEXI — Voice-Controlled Campus Assistant (IoT / Access Control) | VIT *Jan 2023 – May 2023*

Situation: VIT's 200+ acre campus made navigation and information access difficult for students and visitors. Task: Build a voice-controlled assistant with secure access to sensitive queries. Action: Developed an Arduino-based assistant with RFID-backed access control and two-factor authentication, delivering hands-free navigation, event notifications, and faculty look-up. Result: Recognised in VIT's university innovation competition and showcased as a department reference project.

EDUCATION

Master of Technology (M. Tech), Cybersecurity *Expected 2027*

Defence Institute of Advanced Technology (DIAT), Pune — Ministry of Defence, Government of India

Bachelor of Technology (B. Tech), Computer Science Engineering *Graduated 2025*

Vellore Institute of Technology (VIT)

Specialization: Cybersecurity

CERTIFICATIONS

- Introduction to Cybersecurity — Coursera
- Python for Everybody — Coursera / University of Michigan
- Machine Learning and Data Analysis — Udemy
- Arduino and IoT Fundamentals — LinkedIn Learning
- **TryHackMe** — Active Practitioner Profile (tryhackme.com/p/nkeerthivasan): Linux PrivEsc (Rare 2.5%), Bronze League (Rare 5.1%), Hash Cracker (Rare 4.8%), 7-Day Hacking Streak

LANGUAGES

English (Fluent), Tamil (Native), Telugu (Fluent), Hindi (Proficient)